



TB AUDIO PLUG-INS
Team Impulse Impact

USER MANUAL

TB Vocoder (Beta)

Channel Vocoder

Stamps the spectral movement of a voice or rhythm onto a built-in synth or an external carrier, with 8 to 64 bands and five synthesis types.

8 to 64 bands

Internal synth or external carrier

5 synthesis types

119 factory presets



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01 Overview

TB Vocoder is a real-time channel vocoder. It analyses the source in a bank of frequency bands and imprints the movement of each band onto a carrier, so the rhythm and articulation of the source stay recognisable while the tone becomes synthetic.

In SELF mode no second track is needed: the built-in synth is the carrier, so you can load TB Vocoder on a vocal or rhythmic source and start immediately. You can also feed an external carrier through the sidechain input. Choose 8 to 64 analysis bands, set the active frequency range, tune and colour the synth, and shape the response, whitening, width, and sibilance.

Key features

- Filterbank of 8 to 64 bands with an adjustable LOW and HIGH range
- Built-in SELF synth with up to 100 detuned voices, tuning, and a saw-to-pulse COLOR
- External carrier through the sidechain input, with automatic fallback to SELF
- Five synthesis types: VOCODE, WHISPER, MORPH, RING, and INVERT
- SIBILANCE lets consonants read through the synth
- Four-band PRESSURE compression and a TONE tilt after cross-synthesis
- Resynthesis spectrum display with SOURCE, CARRIER, and OUTPUT meters
- 119 factory presets in 12 banks, plus Init and user presets

Signal flow



About this manual

This manual describes TB Vocoder 2.0.0 for Windows (VST3), which is published as a beta. The current macOS build is version 1.0.3; it uses a different control set, including MIDI carrier input, and differs from the descriptions here.

02 Quick start

- 1 Insert TB Vocoder on a vocal or another rhythmic source. The internal synth is the carrier (**SIDECHAIN** shows SELF).
- 2 Play the track. Choose a preset **1**, or start from Init and set **BANDS** **4**: fewer bands sound retro, more bands are more intelligible.
- 3 Set the synth pitch with **TUNE** and its colour with **COLOR** **3**.
- 4 Raise **SIBILANCE** if consonants disappear.
- 5 Shape the response with **ATTACK** and **RELEASE** **4**, and remove background noise with **FLOOR** **5**.
- 6 Blend with **MIX** and set the level with **OUTPUT** **6**.

03 Interface



TB Vocoder 2.0 interface.

- 1 **Preset bar** — previous and next preset, the preset list, Save, Undo, and Redo.
- 2 **Display** — the signal path, band count, and resynthesis spectrum, with SOURCE, CARRIER, and OUTPUT meters.
- 3 **CARRIER** — SIDECHAIN, TUNE, COLOR, COUNT, RESO, and SIBILANCE.
- 4 **FILTERBANK** — BANDS, DETAIL, LOW, HIGH, ATTACK, and RELEASE.
- 5 **IMPRINT** — TYPE, IMPRINT, WHITEN, FLOOR, SHIFT, and WARP.
- 6 **OUTPUT** — PRESSURE, TONE, WIDTH, MIX, and OUTPUT.
- 7 **TYPE** — VOCODE, WHISPER, MORPH, RING, or INVERT.

04 Controls

Carrier

Control	Range · default	What it does
SIDECHAIN	SELF, EXTERNAL · SELF	SELF uses the internal synth as the carrier. EXTERNAL uses the plug-in's Carrier (sidechain) input while enabled, and falls back to SELF when that input is unavailable.
TUNE	-24 to +24 st · 0 st	Transposes the internal synth carrier in semitones; 0 st is the original pitch, A. No effect on an external carrier.
COLOR	0 to 100% · 30%	Morphs the internal synth from a full sawtooth to a narrowing pulse. No effect on an external carrier.
COUNT	2 to 100 · 3	Number of detuned voices in the internal synth.
RESO	0 to 100% · 50%	Width of the synthesis bands. Low is wide and blurred; high is narrow and singing.
SIBILANCE	0 to 100% · 50%	Swaps the carrier for noise while the source is unvoiced, so consonants read through the synth.

Filterbank

Control	Range · default	What it does
BANDS	8 to 64 · 32	Filterbank resolution. Few bands are coarse and retro; many bands track formants closely and stay intelligible.
DETAIL	0 to 100% · 65%	Smears the analysis envelopes across neighbouring bands. Low is broad and soft; high keeps fine spectral detail.
LOW	40 to 500 Hz · 90 Hz	Lowest band centre. Raise it to keep the bank off a bass-heavy source; lower it to imprint the low end as well.
HIGH	2 to 18 kHz · 14 kHz	Highest band centre. Above it, the source is not imprinted.
ATTACK	0.1 to 200 ms · 3 ms	How fast each band envelope rises. Short is percussive and articulate; long softens consonant edges.
RELEASE	5 to 2000 ms · 50 ms	How fast each band envelope falls. Short is tight and gated; long smears words into each other.

Imprint

Control	Range · default	What it does
TYPE	VOCODE, WHISPER, MORPH, RING, INVERT · VOCODE	Synthesis method: classic band vocoding, a noise-carrier whisper, a spectral morph, per-band ring modulation, or an inverted imprint.
IMPRINT	0 to 100% · 100%	Morphs from a carrier gated by the source level to the full spectral imprint.

Control	Range · default	What it does
WHITEN	0 to 100% · 75%	0% keeps the carrier's own spectral balance; 100% makes every output band match the source band exactly.
FLOOR	−90 to −30 dB · −70 dB	Stops low-level source noise from opening the spectral imprint.
SHIFT	−24 to +24 st · 0 st	Moves the spectral imprint without retuning the carrier.
WARP	−100 to +100% · 0%	Stretches or compresses the imprint around 1 kHz without pitch detection.

Output

Control	Range · default	What it does
PRESSURE	0 to 100% · 25%	Four-band upward and downward compression after cross-synthesis.
TONE	−100 to +100% · 0%	Tilts the compressed result between weight and brightness.
WIDTH	0 to 200% · 100%	Stereo spread of the processed signal only. 0% is mono, 100% is unchanged.
MIX	0 to 100% · 100%	Blends the recorded source with the transformed synth.
OUTPUT	−18 to +6 dB · 0 dB	Final output level.

Using an external carrier

- 1 In your DAW, route the carrier track (for example a synth pad) to TB Vocoder's sidechain input.
- 2 Set **SIDECHAIN** to EXTERNAL.
- 3 Play both tracks. The CARRIER meter shows the carrier level. TUNE, COLOR, and COUNT have no effect on an external carrier.

05 Presets

Step through presets with the arrows, or open the list to choose a factory or user preset. Save stores the complete state as a .tbvocoderpreset file. The name shows CUSTOM after you change a setting. Undo and Redo step through recent changes.

The factory set has 119 presets in 12 banks: Essentials, Robots, Whispers, Morph & Formant, Ring & Metal, Invert & Hollow, Low & Chant, Rhythmic, Wide & Cloud, OTT & Squash, Vocoder Resynth, and Carrier Shapes. Examples include **Classic Talk Box**, **Robot Choir**, **Air Whisper**, **Crystal Formant**, **100 Voice Cloud**, **Final Boss Vocoder**, and the band-count series from **Vintage Eight** to **Surgical Sixty Four**.

Presets saved before version 2.0 cannot be loaded, because they use a filterbank and TYPE list that no longer exist.

06 Recipes

Starting points for common jobs. Adjust them by ear.

Classic robot voice

Clear, intelligible vocoded vocals.

Preset Classic Talk Box or Robot Choir

BANDS 24 to 32

SIBILANCE 50 to 70%

- 1 Set TUNE to the song key so the carrier sits in tune.
- 2 Raise SIBILANCE until consonants read clearly.

Retro low-band vocoder

A grainy, vintage sound.

Preset Vintage Eight or Analog Twelve

BANDS 8 to 12

- 1 Fewer bands are less intelligible; keep the vocal part simple.

Rhythmic synth from drums

Imprint a drum loop's rhythm onto a synth.

Preset Percussive Print or Tight Gate

ATTACK Short

RELEASE Short

- 1 Use a pad as an external carrier for chords.

Whisper texture

An airy, breathy layer.

TYPE WHISPER

Preset Air Whisper or Ghost Air

MIX 40 to 70%

- 1 Blend under the dry vocal for atmosphere.

07 Tips and troubleshooting

Symptom	What to check
Words are hard to understand.	Raise BANDS and SIBILANCE, raise DETAIL, or shorten ATTACK.
Background noise opens the vocoder.	Raise FLOOR so low-level noise does not open the imprint.
The bass of the source muddies the result.	Raise LOW to keep the bank off the low end.
The external carrier is not heard.	Check the sidechain routing in your DAW and set SIDECHAIN to EXTERNAL; without a carrier signal the plug-in falls back to SELF.
The carrier is out of tune with the song.	Adjust TUNE (internal synth only) or the external carrier's pitch.
An older preset does not load.	Presets made before version 2.0 are not compatible.

08 Parameter reference

Panel label	Host parameter	Range	Default
CARRIER			
SIDECHAIN	Sidechain	Off (SELF), On (EXTERNAL)	Off

Panel label	Host parameter	Range	Default
TUNE	Carrier Tune	−24 to +24 st	0 st
COLOR	Color	0 to 100%	30%
COUNT	Carrier Count	2 to 100	3
RESO	Carrier Resonance	0 to 100%	50%
SIBILANCE	Sibilance	0 to 100%	50%
FILTERBANK			
BANDS	Bands	8 to 64	32
DETAIL	Detail	0 to 100%	65%
LOW	Range Low	40 to 500 Hz	90 Hz
HIGH	Range High	2 to 18 kHz	14 kHz
ATTACK	Attack	0.1 to 200 ms	3 ms
RELEASE	Release	5 to 2000 ms	50 ms
IMPRINT			
TYPE	Type	Vocoder, Whisper, Morph, Ring Mod, Invert	Vocoder
IMPRINT	Imprint	0 to 100%	100%
WHITEN	Whiten	0 to 100%	75%
FLOOR	Floor	−90 to −30 dB	−70 dB
SHIFT	Shift	−24 to +24 st	0 st
WARP	Warp	−100 to +100%	0%
OUTPUT			
PRESSURE	Pressure	0 to 100%	25%
TONE	Tone	−100 to +100%	0%
WIDTH	Width	0 to 200%	100%
MIX	Mix	0 to 100%	100%
OUTPUT	Output	−18 to +6 dB	0 dB

09 Version history

Version	Changes
2.0.0 · 2026-08-09	Rebuilt the vocoder around a configurable multi-band filterbank with LOW and HIGH range, attack, release, width, and resynthesis controls. Redesigned the interface into four focused panels and introduced a new state format for the 23-control engine.

10 Support and license

Report problems or request features at github.com/TaebeumKim/TB_Audio-Plugins/issues. Include the plug-in version, your DAW, and your operating system.

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